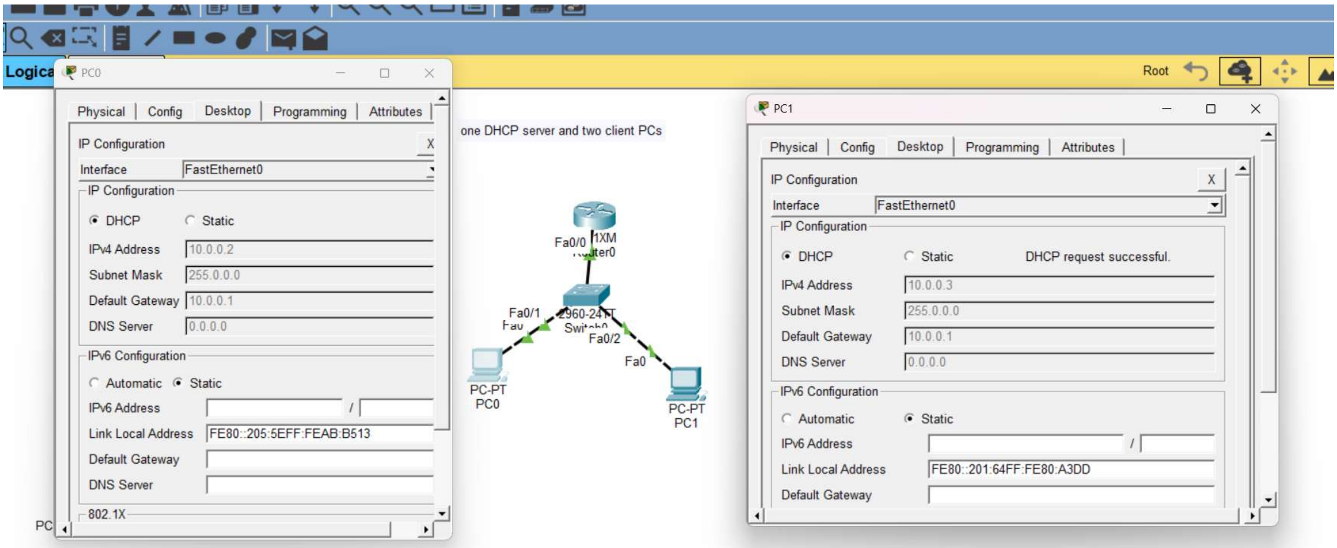
Day 2. DHCP configuration Theory

Task 1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.



Task 2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint: Use arrows to show the direction of each message between client and server.

STAGE 1: DISCOVER (D)

- **Action:** The client boots up without an IP and sends a general broadcast query to find any active DHCP server on the local switch segment.

STAGE 2: OFFER (O)

- **Action:** The router receives the request, reserves an available IP (`10.0.0.2`) from its active pool scope, and sends an offer back to the client MAC.

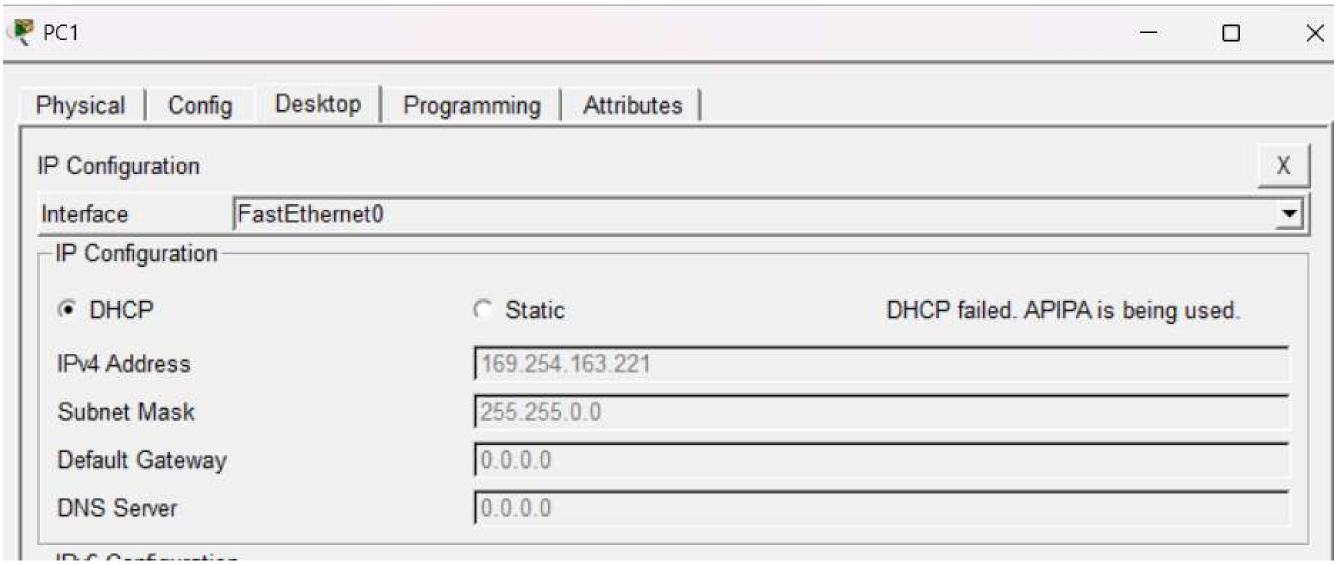
STAGE 3: REQUEST (R)

- **Action:** The client broadcasts its formal request to claim `10.0.0.2`, notifying all network listening servers which offer it chose.

STAGE 4: ACKNOWLEDGE (A)

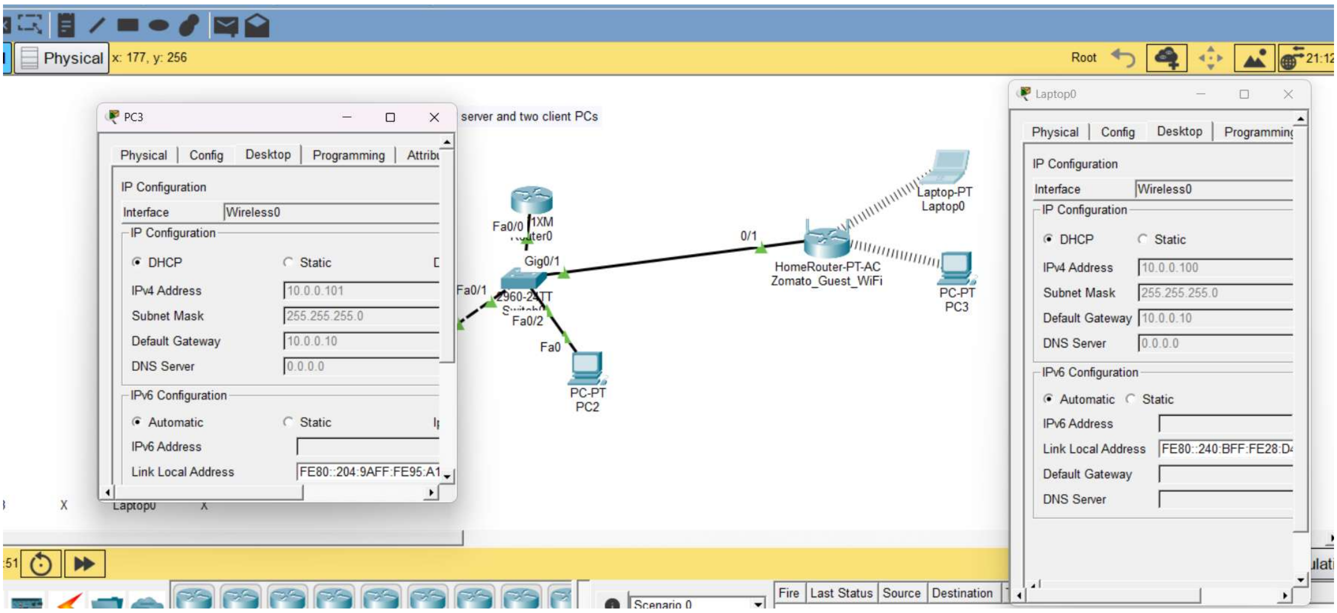
- **Action:** The DHCP server locks the IP binding in its database and sends an acknowledgment packet containing subnet mask, default gateway, and DNS configuration.

Task 3.Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.



Only PC0 succeeds because the router's pool contains a single usable address (`10.0.0.2`), which PC0 leases first. When PC1 sends a `DHCP DISCOVER` request, the server experiences scope exhaustion and cannot offer an address, causing PC1 to default to an APIPA address (`169.254.163.221`)

Task 4.Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.



In a public place like a Zomato-partnered restaurant, hundreds of unique guest devices constantly join and leave the network. DHCP automates seamless plug-and-play IP address allocation and recycles expired IP leases, eliminating the manual overhead of configuring static IPs for every customer.